

STAT 408 Lab 7: Logicals

Feb 26, 2026

You can collaborate and submit this lab in pairs or individually. If you work in a pair, please submit it as a group (pair) assignment instead of individually.

Create a new Quarto document titled “STAT 408 Lab 7: Logicals” and put your name and some sensible date (the due date or today’s date, for instance). Throughout your assignment, write your code with good style, organization, and comments. Use your document to do the following, and use the same headings as shown below.

If there are additional things you think are interesting to do with the data, feel free to try them out! If you get stuck on things or have questions outside of class, please feel free to post your questions on the Lab 7 Canvas discussion board.

Problem 1: Read in the survey data

Read in the survey data your group collected in class. (If you were not in class, pick one of the datasets posted on the Canvas discussion board.)

- Show your code for reading in the data, but do not show any code output.
- Make sure that you read in the variables as reasonable data types (for instance, numbers should not be handled as character/text data).
- Give your variables short, informative names with a consistent style.
- Make sure that missing values are read in as `NA` (not a text “NA” or other values like “N/A” or -99).

Problem 2: Explore the survey data

Explore the survey data in R.

- Show your code for this section, and summarize the results in markdown text. Only show code output if it summarizes and presents your answers, not if you have to hunt through a bunch of output to find the answers.
- Which variables have a limited set of fixed possible answers? Use your code to compute how many people or what proportion of respondents gave each answer, and summarize the answers in a table or a sentence.

- Which variables are numeric? Comment on the distributions of these variables: are they spread out or concentrated around a particular value, are there multiple modes/common values, are there outliers?
- Do any of the variables have missing data? Which variables, and how much missing data? Do they seem to follow any pattern (are there certain questions people tended to avoid answering, or questions that tend to be both be missing if one is missing, etc.)? For this question, make at least one table summarizing the missing data in some way that makes sense for your dataset.
- Are there any outliers? Any values you think might be incorrect? Briefly summarize.

Problem 3: Manipulate the data

Let's get some practice with logical comparisons and transformations.

- Show your code for this section, and summarize the results in markdown text. Only show code output if it summarizes and presents your answers, not if you have to hunt through a bunch of output to find the answers.
- Create a new column that equals TRUE if another column is greater than some threshold and FALSE otherwise. Give the new column a sensible name, and describe in a sentence of markdown text what it does.
- You probably have a column that has some typos in it, or multiple values that ultimately mean the same thing. Create a new column that condenses these values into the same value. For example, if you have a column called `country` and you think the values "USA" and "United States" in that column mean the same thing, make a new column (or modify the `country` column) so that the value is the same for rows with `country` "USA" and rows with `country` "United States".
- Create a new column that equals at least four different possible values (could be numeric or text) depending on the value of one or more other columns. (Hint: use `case_when`.) Give the new column a sensible name, and describe in a sentence of markdown text what it does.